

Conjecture 00000000537 is false:
 K_2 has depth $R/I(K_2) = 1 < 2 = \nu(K_2) + 1$

Jackmeson1

1 The conjecture

The statement (English text of `conjectures/00000000537.md`):

Definition: The depth of $R/I(G)$ versus the matching number $\nu(G)$. Conjecture: For bipartite graphs, depth is at least $\nu + 1$ (partially known; the complete statement is the residual direction, here taken as a new strictly-inequality class: the chordal bipartite equality characterization).

The Chinese text has the same main clause: for bipartite graphs, depth $\geq \nu + 1$.

Reading. We read the main clause literally: for every finite bipartite graph G and every field k , depth $R/I(G) \geq \nu(G) + 1$, where $R = k[x_v : v \in V(G)]$ and $I(G)$ is the edge ideal. We refute this clause; the parenthetical remark does not weaken it, and we make no claim about the parenthetical “equality characterization” (which the text does not state precisely). The counterexample K_2 works over every field, so no choice of the field rescues the clause.

2 Definitions (as formalized)

Let k be a field, V a type of vertices and G a simple graph on V . Let $R = k[x_v : v \in V]$.

Definition 1. The *edge ideal* is $I(G) = (x_u x_v : uv \in E(G)) \subseteq R$ (Lean: `C537.edgeIdeal`). The *homogeneous maximal ideal* is $\mathfrak{m} = (x_v : v \in V)$ (Lean: `C537.irrelevantIdeal`).

Definition 2. A list $r_1, \dots, r_n \in R$ is an *M -regular sequence* (Mathlib: `RingTheory.Sequence.IsRegular`) if each r_i is a nonzerodivisor on $M/(r_1, \dots, r_{i-1})M$ and $M \neq (r_1, \dots, r_n)M$. The *depth* of $R/I(G)$ is

$$\text{depth } R/I(G) = \sup\{n : \text{some } R/I(G)\text{-regular } r_1, \dots, r_n \text{ has all } r_i \in \mathfrak{m}\}$$

(Lean: `C537.depth`), i.e. the depth with respect to the homogeneous maximal ideal.

Definition 3. A *matching* of G is a subgraph in which every vertex of the subgraph lies on exactly one of its edges (Mathlib: `SimpleGraph.Subgraph.IsMatching`). The *matching number* $\nu(G)$ is the supremum, in $\mathbb{N} \cup \{\infty\}$, of the number of edges of a matching (Lean: `C537.matchingNumber`). A graph is *bipartite* if it is 2-colourable (Mathlib: `SimpleGraph.IsBipartite`).

The graph K_2 is the complete graph on two vertices $\{0, 1\}$ (one edge). It is bipartite, and $I(K_2) = (xy) \subseteq k[x, y]$ with $x = x_0$, $y = x_1$.

3 The proof

Lemma 4 (`matchingNumber_K2`). $\nu(K_2) = 1$.

Proof. The single edge is a matching. Every edge of a subgraph of K_2 is the edge $\{0,1\}$, so a matching has at most one edge. $\square$

Let $\varphi, \psi : R \rightarrow R$ be the k -algebra maps with $\varphi(x) = x$, $\varphi(y) = 0$ and $\psi(x) = 0$, $\psi(y) = y$ (Lean: `phiY`, `psiX`). For every $f \in R$ we have $f - \varphi(f) \in (y)$ and $f - \psi(f) \in (x)$ (`sub_phiY_mem`, `sub_psiX_mem`, both from the general `sub_mem_of_gen`), $\varphi \circ \varphi = \varphi$, $\psi \circ \psi = \psi$, and $\psi(\varphi(f)) = 0$ for $f \in \mathfrak{m}$. Also $x \notin (xy)$ and $y \notin (xy)$ (evaluate at $(1,0)$, resp. $(0,1)$).

Lemma 5 (`no_regular_pair`). *Let $M = k[x,y]/(xy)$. There are no $f_1, f_2 \in \mathfrak{m}$ such that f_1 is a nonzerodivisor on M and f_2 is a nonzerodivisor on M/f_1M . Hence no M -regular sequence in $\mathfrak{m}$ has length ≥ 2 .*

Proof. Put $u = \varphi(f_1) = f_1(x, 0)$.

$u \neq 0$ and $\psi(f_1) \neq 0$. If $u = 0$ then $f_1 = f_1 - \varphi(f_1) \in (y)$, so $f_1x \in (xy)$, i.e. $f_1 \cdot \bar{x} = 0$ in M with $\bar{x} \neq 0$, contradicting regularity. Symmetrically $\psi(f_1) \neq 0$.

$\mathfrak{m}\bar{u} \subseteq f_1M$. Write $f_1 - u = yc$. Then $xu = xf_1 - xyc \equiv f_1 \cdot x$ modulo (xy) . Also $u = u - \psi(u) \in (x)$ since $\psi(u) = \psi(\varphi(f_1)) = 0$, so $yu \in (xy)$. Thus $x\bar{u}, y\bar{u} \in f_1M$, and hence $r\bar{u} \in f_1M$ for all $r \in \mathfrak{m}$.

$\bar{u} \notin f_1M$. Suppose $u - f_1g = xyh$. Applying φ gives $u = u\varphi(g)$, so $\varphi(g) = 1$ as R is a domain and $u \neq 0$. Applying ψ gives $0 = \psi(f_1)\psi(g)$, so $\psi(g) = 0$. Then $1 = \psi(\varphi(g)) = \varphi(\psi(g)) = 0$, a contradiction.

Finally $f_2 \in \mathfrak{m}$ gives $f_2\bar{u} \in f_1M$, so f_2 kills the nonzero class of $\bar{u}$ in M/f_1M , contradicting that f_2 is a nonzerodivisor there. $\square$

Theorem 6 (`depth_K2_le`, `K2_counterexample`). *For every field k : K_2 is bipartite, $\nu(K_2) = 1$, and $\text{depth } R/I(K_2) \leq 1 < 2 = \nu(K_2) + 1$.*

Proof. $I(K_2) = (xy)$ (`edgeIdeal_K2`); an $R/I(K_2)$ -regular sequence in $\mathfrak{m}$ of length at least 2 would contradict the previous lemma. $\square$

Theorem 7 (`conjecture_537_false`). *It is not the case that every finite bipartite graph G satisfies $\text{depth } R/I(G) \geq \nu(G) + 1$ over every field.*

Remark 8. In fact $\text{depth } k[x,y]/(xy) = 1$ (the element $x+y$ is regular), but only the upper bound is needed and only the upper bound is formalized.

4 Lean formalization

File `lean/Conjecture537/Basic.lean` (Lean 4, Mathlib v4.33.1, only `import Mathlib`). Main statement (ASCII transliteration of the Lean source, which writes the Unicode symbols for `Not`, `forall`, `->`, `<=` and `/\`):

```
theorem conjecture_537_false :
  Not (forall (V : Type) [Fintype V] (G : SimpleGraph V), G.IsBipartite ->
    forall (k : Type) [Field k], matchingNumber G + 1 <= depth k G)
```

and, for every field k ,

```

theorem K2_counterexample (k : Type*) [Field k] :
  K2.IsBipartite /\ matchingNumber K2 = 1 /\ depth k K2 <= 1 /\
    depth k K2 < matchingNumber K2 + 1

```

Here `matchingNumber` and `depth` are `C537.matchingNumber` and `C537.depth`, valued in $\mathbb{N} \cup \{\infty\}$ (`ENat`). `#print axioms` for both theorems: `[propext, Classical.choice, Quot.sound]`. No sorry, no `native_decide`, no added axioms.